Blockchain Based Car-Sharing Platform

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***Abstract*—The car-sharing market is constantly growing and recently it has become even more popular than car ownership. However, classic car-sharing system is based on a centralized database server which can often lead to hacker attacks or password leaks. Moreover, in a classic car-sharing system, the owners of the cars can misuse customers' data. As seen nowadays from a lot of use cases, the best solution to these problematic issues is to use blockchain technology. Blockchain as decentralized, immutable, public ledger provides the customers with security that is impossible to tamper. The aim of the proposed solution is to create and implement peer-to-peer short term car-sharing application based on blockchain technology and smart contracts. For the implementation of smart contracts, Solidity programming language is used. Solidity works with Ethereum blockchain. The key novelty of the article is introducing a peer-to-peer car sharing service without a central authority, what reflects a decrease of costs and increase of data transparency in that system. Also token based solution gives us ability to cover business-to-business (B2B) and business-to-customer (B2C) use cases.**

***Keywords*—Blockchain; Car sharing; Asset sharing; Ethereum; ERC-721; Smart contract**

I. Introduction

An increasingly popular approach is the sharing of cars. While many people would not like to share a car with a foreigner, car renting solutions such as car2go, where companies share they owned cars, are a great opportunity in cities. Gas powered vehicles are the main source of pollution and therefore the trend in many countries is to reduce the number of cars on roads [1]. Car-sharing can solve also this problem by providing us with the car when we need it. According to statistic car is parked for 96% of the time [2]. Furthermore, we are not forced to buy a car what significantly reduces our living and maintenance costs. Smart ideas side by side with mobility are calling for unconventional, probably even unfriendly advances [3]. That is why we want to propose a car sharing platform using blockchain technology. This idea is not new and we are not the first ones to write about it. We came to it when we were thinking about possible use cases of the blockchain in real environment. When looking for ways to share vehicles online, we found that many large companies and even startups are already working on specific solutions. There are even startups specialized in this particular problem.

One of the biggest peer-to-peer car-sharing platforms is Getaround [4]. That platform allows users to rent a car for a small period of time, starting from one hour. Users are able to earn some money from sharing their cars when they do not need them.

Unfortunately, providing this kind of service to users brings a lot of security flaws; first of all is overall stability of service. Using DDoS attacks on centralized server would lead into collapse of the whole service. In comparison, when data are distributed and processed over whole network of nodes like blockchain, it is practically impossible to tamper the service [5].

In addition, traditional data servers provide us CRUD interface so we are able to create, read, update and delete data. This suggests that some people might change users' data without further notification of the users. On the contrary, blockchain as immutable, public ledger allows us only to create and read data. If we want to update some information in blockchain, we have to notify every single user in the network that we have changed data. As a matter of course, it is also impossible to delete data from blockchain. It can be concluded that blockchain as unchangeable method increases transparency and ensures data protection [6]. Moreover, using public ledger allows us to cut fees which are needed for paying car sharing employees, such as system administrators.

All of these points mentioned above have become the core motivation to create a decentralized peer-to-peer application built on top of Ethereum blockchain by using Solidity language for smart contracts and ReactJS for client side of the application. As a connector to Ethereum network we are using Web3. However, as mentioned before, we were not the first who had the idea to implement peer-to-peer car-sharing platform. There are projects such as HireGo [7], DAV [8] or Helbiz [9], but most of these project are in a stage of very early prototype.

The following sections of this article are organized as follows: Section II shows current state of the art solutions in domain of car-sharing with blockchain. Section III describes design architecture of our decentralized car-sharing project with focus on our tokenized architecture. Evaluation and Testing to proof our solution are in Section V. Section VI sums up the article with results and we also discuss future work there.

II. Related Work

In this section we are covering known solutions which are for now considered state of the art.

First application is HireGo. It is built on Ethereum using smart contracts as a distributor of virtual fungible tokens (standard ERC-20) [10], called HGO. When we want to rent

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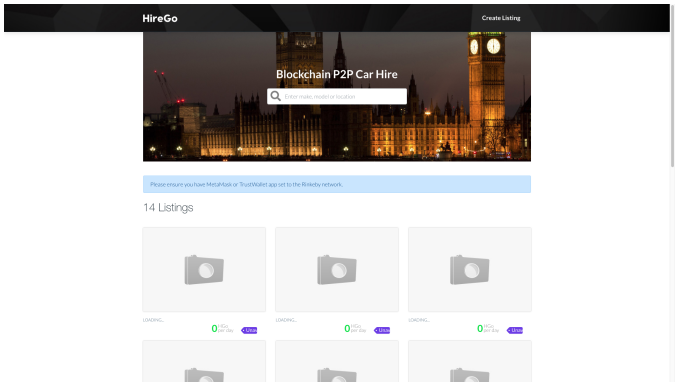

Figure 1. Demo application of HireGo

a car using HireGo, we need to exchange ETH to HGO tokens. HireGo's market place consists of three contracts. First contract provides HGO tokens, second one provides ERC-721 car tokens and the last one provides rental contract. This rental contract acts as an escrow and it is responsible for Vehicle token between two parties [7]. In other words, when we rent a car, we send HGO tokens to rental smart contract which locks car token and HGO tokens for selected period. Then the contract manages the transfer of the vehicle token from owner to us. The rental solution also brings big security issue. If we rent a car for a day trip and then we loose the phone with Ethereum address, the car will be stuck for the rest of time and the owner is unable to help us because he/she does not own that vehicle at the time of renting. In the Figure 1 we can see alpha version of decentralized application made by HireGo. Unfortunately, the developers allowed us only the ability to create car for rent in this version.

Next application we will introduce is Helbiz. Helbiz's philosophy is to create and integrate transport ecosystem, where we are rewarded for using this system. This project is using well-known implementation of blockchain car-sharing platform by providing ERC-20 tokens called HBZ. In contrast to HireGo, we can also earn tokens by providing our data about driving to insurance companies [9]. Helbiz is the only project which has created a mobile application for renting electric scooters as we see in the Figure 2. When we want to rent a scooter we simply find a free one in the application and then we scan QR code, and pay for the service with HBZ tokens. If we do not have enough tokens, the app offers to buy HBZ tokens by using credit card. Unfortunately, we are not able to test this solution because all of the scooters are located in Italy.

Next project we will look at is DAV. As opposed to other projects the main goal is to connect autonomous vehicles (cars, trucks and drones) with users. For communication purposes we use DAV token, which also serves as access token. Services in DAV network can be bought with virtual tokens based on Ethereum. It shall not be mistaken with DAV token. If we are owning vehicle or charging station which we provide to other users in network we are rewarded with DAV tokens [8]. Autonomous vehicles in network automatically interact with

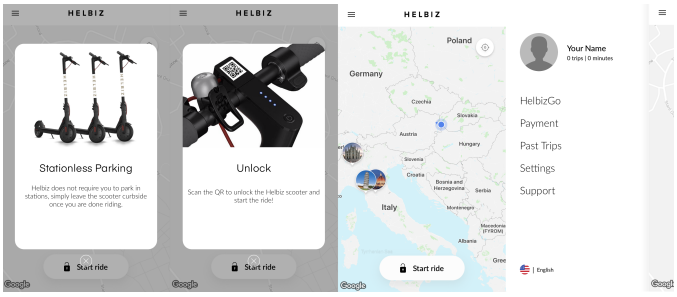

Figure 2. Demo application for scooter rental by HBZ

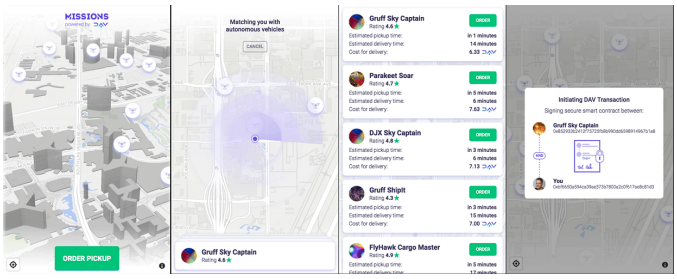

Figure 3. Demo application for drone management

environment and also can fulfill user's commands. As we see in the Figure 3 with demo application we can simply interact with drone.

In contrast to other projects FFQuest's main strength is usage of their own chain called Distributed FFQ Ledger based on Ethereum Virtual Machine. This ledger will pass all transaction details between car companies, drivers and customers such as payment conditions, assets, availability, and reservations by using their own utility token, type ERC-20 called FFQ. This project uses central backend to mirror Distributed FFQ Ledger on a server and also for storing videos and images [11].

The last known project is WONO. WONO used different approach than the other projects, by providing a decentralized market for renting services such as houses, cars and working man days. Their solution is based on the set of smart contracts that work as a brigde between Ethereum main net and their blockchain. WONO blockchain is using Proof-of-Stake algorithm to ensure consensus in the network. They store user's profile data, represented as JSON, in the decentralized IPFS file system. Data properties which contain confidential information are encrypted on a client's app side by a public key of the User's address and can be decrypted only by private key [12].

III. Design

As we mentioned earlier, there are some projects trying to implement peer-to-peer car-sharing platform. Thus, we will attempt to compare our project with those solutions' specifications. In contrast to related work and what we have learnt, we try to build our solution with two different tokens based on non-fungible token type ERC-721 and one fungible token type ERC-20. We can simply describe ERC-721 [13]

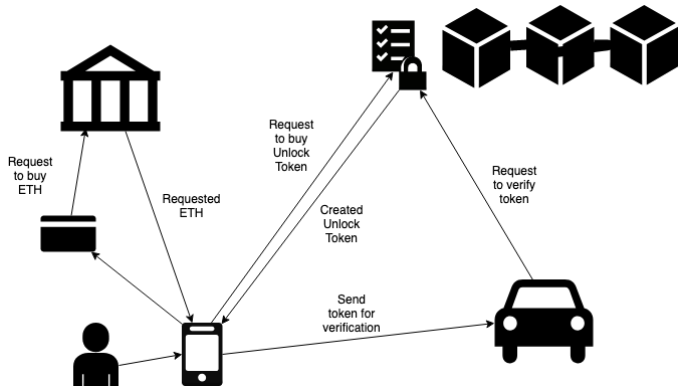

Figure 4. Graphic representation of our solution

as a smart contract which holds list of unique assets. These assets are represented as non-fungible tokens. First token type represents car asset and the second one represents Unlock token for car token. This solution will allow us to scale the app more efficiently. The car owner, by the very nature of blockchain, would have full control over who gets access to his or her information, and the same goes for the person borrowing the vehicle. Without a key to piece together the distributed data and then decrypt it, it is far harder for a third party to hack either side's personal information. We applied ERC-20 token to reward user for using our car-sharing application. After user collects particular amount of tokens, he can apply for a discount to renting a vehicle. Users are then more motivated to use our service if they can get it cheaper. Also we managed to save car token's image to IPFS, which is decentralized storage, to reduce costs of ethereum transactions. Storing IPFS URI is much more effective and cheaper than storing image on the blockchain network. As a gateway to IPFS network, we decided to use Pinata instead of hosting our own, because of really simple implementation to JavaScript, fair pricing and overall speed of service [14]. At the beginning of this project we were thinking how to provide as wide scalability as possible. Moreover, we bore in mind that providing users positive experience will convince other users to use our application. In the Figure 4 we can see the user interaction with the system. When we want to rent a car, we just select the car, length of the renting period, and pay with our preferred currency (USD/EUR). Application in the background exchanges money to cryptocurrency (ETH in our case) and after that executed smart contract will provide us Unlock Token to unlock the car. In our project, we do not force users to buy ETH and convert them into other fungible tokens. Instead we automatically convert real currency to ETH exactly in time when user needs it.

We use an IoT device in a car, which is waiting for unlock token. The IoT device can be managed by any standard means or also by blockchain as introduced e.g. in [15], [16]. If vehicle receives unlock token, it will verify its validity with blockchain network. After validation being successful, the car is ready to be taken for a ride. We take on two scenarios which help us to understand token solution better.

The first scenario is client-to-client solution. The user provides a car for renting. That means he is the owner of unique non-fungible (NF) car token and also owner of the NF master unlock token that could unlock the car anytime. When the other user wants to rent a car for a period of time, he will receive other NF unlock token for the particular car. The metadata in unlock token describes details about renting. When car owner decides to sell the car, car token and master unlock token just simply change the owner. Also car can have more master unlock tokens (something like spare car keys).

The second scenario describes usage of this kind of design for business-to-customer (B2C) and business-to-business (B2B) solution. Our platform can also provide solution for business car rental services such as ZipCar. Business service will provide car fleet, which can be rented by users and other businesses. Then, for each car many master unlock tokens can be created (e.g. for CEO, for person responsible for repairs, etc.), and also many rental unlock tokens can be created (e.g. company rents a car for a long period of time to many employees).

IV. Test scenarios

After implementation phase we wanted to make sure that our application is working, so we created two test scenarios. The main goal was to test overall speed, stability and functionality of our proposed solution.

Our testing environment consisted of a MacBook Pro with MacOS Mojave (10.14.3), Chrome web browser with enabled JavaScript and installed MetaMask plugin. MetaMask provides simple interface for communication with ethereum network. Also for testing purposes we did not deploy smart contract to main or any test network. Instead we created our local ethereum network using Ganache. Ganache is a one click personal Ethereum blockchain.

V. Results

First scenario covers creating new car available for renting and also creating unlock token for the user. Every transaction on blockchain network needs to be payed by user. The motivation behind this test was to get actual price that user needs to pay in order to create a car available for renting. Also we can consider this amount of money as initial investment for user to list the car. The process of creating new car is pretty straightforward. We simply open web dApp, log in to the network via MetaMask, fill out the form with name, description and picture of the car. The process is simple, user friendly and the whole scenario takes only 15 seconds to complete. Calling create car token function costs user about 0.014485 ETH.

In second scenario we tested a time and responsiveness of our application. Our test covers case from opening application to open details with car and receive token for unlocking. We wanted to find out how long does it rent and unlock car. This test scenario is crucial for our application, because one of our motivation is to provide smooth user experience. If the test scenario would take too long, user can become impatient and

TABLE I. Test results of time needed for renting

Test nr.	Time (s)
Test 1	13.41
Test 2	11.11
Test 3	10.35
Test 4	12.9
Test 5	14.78
Test 6	10.40
Test 7	12.73
Test 8	13.30
Test 9	11.57
Test 10	10.91
Average	12.146

could leave the app very soon. However, based on results, using the application for renting a car is fast and comfortable. As we see in the Table I, the average time for purchasing a rent token is 12.146 s.

VI. Conclusions and Future Work

Our primary focus was to create a peer-to-peer car-sharing decentralized application. We were able to fix problems that centralized services were facing such as denial of service and data mutability by providing decentralized and immutable blockchain technology. Our solution adds security to the whole ecosystem, firstly by utilizing blockchain, and secondly by using two different types of ERC-721. Again, using the example of hijacking a car, blockchain can make it much harder for a hacker to gain control, because it is only being given a digital "key" to access the vehicle's systems. If this key is distributed around the internet, and accessible only with its owner's permission, it is immeasurably harder to hack than a cybercriminal being faced with only hacking their way into a single server. Tokenization of car keys helped us to cover many use cases connected with car-sharing services, such as preventing car from partial deadlock. As we mentioned before, current implementation of our decentralized application covers major scenarios such as creating the car token with major unlock token and providing tokenized renting solution. According to test scenarios, creating a car available for renting seems to be pretty straightforward. Additionally, users are able to rent a car in average 12.146 seconds, which brings the same user experience as using centralized car-sharing solution.

In near future we are going to add different Unlock token types, because actual implementation allows us only to create master unlock token. Moreover, our design proposed automatic exchange currency to cryptocurrency, but we did not find simple exchange market for this kind of use case. Additional improvement proposal is to provide seamless user experience with log-in to our application which means that users are not forced to remember their private keys and they will be able to use their email address instead of it.

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